4.6 Application of rates and equilibrium

Students will be assessed on their ability to:

a demonstrate an understanding of how, if at all, and why a change in temperature, pressure or the presence of a catalyst affects the equilibrium constant and the equilibrium composition and recall the effects of changes of temperature and pressure on rate, eg the thermal decomposition of ammonium chloride, or the effect of temperature and pressure changes in the system $2NO_2 \rightleftharpoons N_2O_4$

b use information on enthalpy change and entropy to justify the conditions used to obtain economic yields in industrial processes, and understand that in reality industrial processes cannot be in equilibrium since the products are removed, eg in the Haber process temperature affects the equilibrium yield and rate whereas pressure affects only the equilibrium yield (knowledge of industrial conditions are not required)

c demonstrate an understanding of the steps taken in industry to maximise the atom economy of the process, eg recycling unreacted reagents or using an alternative reaction